For the 480 V side of the system, I first identified the electrical requirements of each major component and then looked through manufacturer datasheets to determine how they would realistically connect and what ratings the surrounding components would need. The system starts with the provided **480 V three-phase generator**, followed by an **Eaton DH362UGK three-pole manual disconnect**. I chose this because its 60 A, 600 VAC rating is sufficient for the system and it allows all three phases to be physically isolated for maintenance or an emergency. From there, an **Eaton PDB321-3 three-phase distribution block** splits the supply into the motor and indicator branches. I chose it because its datasheet specifies three poles, a 600 VAC rating, and more than enough current capacity for the system.

For the motor branch, I used **three 40 A Class J fuses** before the VFD for short-circuit and overcurrent protection. Rather than choosing 40 A arbitrarily, I got this directly from the datasheet for the **Invertek ODE-3-340240-3F42**, which specifies 27.5 A continuous input current and 40 A input protection. I selected this VFD because it is rated for **11 kW / 15 HP, 380–480 V three-phase input, and 24 A output**, making it appropriately sized for the motor in this problem. The VFD is the main control component of this branch; it takes the fixed-frequency three-phase supply and creates a controlled three-phase output so the motor can operate at three preset speeds selected through the VFD's digital inputs. I also researched adding an **input choke** before the VFD. Invertek recommends one for conditions such as phase imbalance, brownouts, or low supply impedance/high available fault current, but because its necessity depends on the actual generator and installation, I treated it as an additional recommended peripheral rather than assuming it was required.

The other branch powers the **480 V AC indicator light**, which provides a visual indication that the high-voltage system is energized. Since the generator has no neutral, the 480 V indicator is connected between two phases, such as L1 and L2, with its own appropriately sized branch protection; protective earth is never used as a substitute for neutral. PE would instead be bonded to components such as the **VFD chassis, motor frame, and conductive enclosure** to provide a safe fault-current path. Overall, I used the component datasheets not just to find parts that said “480 V,” but to make sure their **voltage, current, phase, protection, and connection requirements were compatible with the components around them**.

